Coauthor Analysis: Dimensional Repair Framework

All Remaining Errors, EFT Cleanup, Experimental Redesign,
and Concrete Forward Path

Responding to K. Monette, *Dimensional Repair and Refined Framework*, March 2026

Coauthor-Level Technical Review

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1. Executive Verdict

Executive Verdict

The dimensional fix is real and important: the original framework had a genuine units failure, and introducing E_* resolves it. This is genuine scientific progress.
However, three new errors were introduced in the revision, and one structural problem was revealed that will require significant work to address. The document as it stands should *not* be submitted without the following repairs:

1. **Wrong power of c in the EH action.** The file writes $c^3/(16\pi G)$ in the Einstein-Hilbert Lagrangian. The correct coefficient is $c^4/(16\pi G)$. This is not a typo—it propagates through the dimensional analysis.
2. **Numerical error in the signal calculation.** The claimed $\Delta a \approx 3.6 \times 10^{-6}\,\mathrm{m/s^2}$ is wrong by a factor of ~ 1000. The correct value is $\Delta a \approx 3.6 \times 10^{-9}\,\mathrm{m/s^2}$. The optomechanical conclusions change but do not disappear.
3. **E_* is not a "repair" — it is a new free parameter.** The original framework had one undetermined coupling ($\tilde{\kappa}$); the repaired framework has two ($\tilde{\kappa}$ and E_*). Experiments constrain only the product $\tilde{\kappa} E_* \alpha_{\mathrm{screen}}$. Until E_* is motivated or constrained, the theory has less predictive content than before. This must be stated honestly.
4. **The Cooper-pair constraint.** If $E_* \approx E_P$ (the only choice that makes any experiment viable), then macroscopic superconductors would have effective "entanglement masses" vastly exceeding their actual mass, producing gravitational anomalies that have never been observed. This is potentially the strongest existing bound on the theory and must be addressed explicitly.

Strategic assessment: The dimensional fix points toward a genuinely interesting result if E_* can be derived or constrained to lie near the Planck scale. The correct experimental direction (near-field force sensing over Ramsey interferometry) is right. The framework is not broken, but it is not yet coherent. The Cooper-pair constraint is the single most important issue to resolve.

1 2. What Is Now Solid

Solid

The dimensional fix (Section 2–3 of the white paper) is mathematically correct and scientifically necessary. The original coupling αS_{ent} with $\alpha = \tilde{\kappa} c^4/(8\pi G k_{\mathrm{B}} \ln 2)$ produces $[\alpha S_{\mathrm{ent}}] = \mathrm{K/m^4}$, not $\mathrm{J/m^3}$. The introduction of E_* with $[E_*] = \mathrm{J/bit}$ restores dimensional consistency. This is a genuine correction.

The following structural results are also correct:

1. **External-source action:** Treating $\phi(x) \equiv S_{\mathrm{ent}}(x)$ as a prescribed external source (not varied in the action) is the correct resolution of the earlier logical tension. The action varies only with respect to $g^{\mu\nu}$; conservation of total stress-energy follows from $\nabla^\mu G_{\mu\nu} \equiv 0$ alone.

2. **Observable combination:** The observable $\mathcal{B} = |\tilde{\kappa}_{\mathrm{eff}}|\, \Delta M_{\mathrm{ent}}/R_{\mathrm{eff}}^2$ with $\Delta M_{\mathrm{ent}} = E_* \Delta S_{\mathrm{ent}}$ now carries correct dimensions of energy density ($\mathrm{J/m^3}$). Experiments constrain this combination, not $\tilde{\kappa}$ alone.

3. $1/R^2$ **exterior scaling:** For bounded sources, $\Delta a(R) \propto M_{\rm ent}(<R)/R^2$ is a universal consequence of spherical symmetry and is dimensionally and mathematically correct.

4. **Experimental pivot:** The direction toward near-field force sensing over Ramsey interferometry is correct in principle. The clock phase shift has an extra suppression by c^2 relative to the direct acceleration signal, making force sensors the better primary observable.

5. **RG framing:** Writing $\tilde{\kappa}_{\rm lab} E_{*,\rm lab} = \tilde{\kappa}_{\rm UV} E_{*,\rm UV} (E_{\rm lab}/E_P)^\gamma$ is the correct parameterization. Experiments constrain the infrared product $\tilde{\kappa} E_*$; connecting to $\tilde{\kappa}$ alone requires additional input from OP10.

2 3. What Still Fails or Remains Unclear

2.1 Error 1: Wrong power of c in the Einstein-Hilbert action

Error

Line 118 of `whitepaper-dimensional-fix.tex` (and the corresponding equation in both the PDF and Gemini output) writes:

$$S[g;\phi] = \int d^4x \sqrt{-g} \left[\frac{c^3}{16\pi G} R - \cdots \right].$$

The correct Einstein-Hilbert Lagrangian density is $(c^4/16\pi G)R$, not $(c^3/16\pi G)R$.

Proof. In SI units, the action must have dimensions of $\mathrm{J} \cdot \mathrm{s}$.

$$\begin{aligned}
[d^4x] &= \mathrm{m}^3 \cdot \mathrm{s}, \\
[\sqrt{-g}\, R] &= \mathrm{m}^{-2} \quad (R = \text{Ricci scalar}), \\
\Rightarrow \quad \left[\frac{c^4}{16\pi G}\right] &= \frac{\mathrm{m}^4\,\mathrm{s}^{-4}}{\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}} = \mathrm{kg} \cdot \mathrm{m} \cdot \mathrm{s}^{-2}, \\
\Rightarrow \quad \left[\frac{c^4}{G} \cdot d^4x \cdot R\right] &= \mathrm{kg} \cdot \mathrm{m} \cdot \mathrm{s}^{-2} \times \mathrm{m}^3 \cdot \mathrm{s} \times \mathrm{m}^{-2} = \mathrm{kg} \cdot \mathrm{m}^2 \cdot \mathrm{s}^{-1} = \mathrm{J} \cdot \mathrm{s}. \quad \checkmark
\end{aligned}$$

For $c^3/(16\pi G)$: $[c^3/G] = \mathrm{kg/s}$, giving $[c^3/G \cdot d^4x \cdot R] = \mathrm{kg} \cdot \mathrm{m}$, which is not an action. The c^3 form is therefore dimensionally wrong.

Corrected action:

$$S[g;\phi] = \int d^4x \sqrt{-g} \left[\frac{c^4}{16\pi G} R - \frac{\beta}{2} g^{\mu\nu} (\nabla_\mu \phi)(\nabla_\nu \phi) - \tilde{\kappa} E_* \phi \right]. \tag{1}$$

2.2 Error 2: Numerical error in the signal estimate

Error

The Gemini/AI Studio output claims $\Delta a \approx 3.6 \times 10^{-6}\,\mathrm{m/s^2}$ for the unsuppressed signal ($\tilde{\kappa}_{\rm eff} = -1/4$, $\Delta S_{\rm ent} = 100\,\mathrm{bits}$, $R = 100\,\mu\mathrm{m}$, $E_* = E_P$). This is wrong by a factor of $\sim 10^3$.

Correct calculation. The "entanglement mass" per bit is

$$m_{\rm bit} = \frac{|\tilde{\kappa}| E_*}{c^2} = \frac{(1/4) \times 1.957 \times 10^9\,\mathrm{J}}{(2.998 \times 10^8\,\mathrm{m/s})^2} = 5.44 \times 10^{-9}\,\mathrm{kg} = 5.44\,\mu\mathrm{g/bit}. \tag{2}$$

For $\Delta S_{\text{ent}} = 100\,\text{bits}$ at distance $R = 100\,\mu\text{m}$:

$$M_{\text{ent}} = 100 \times m_{\text{bit}} = 5.44 \times 10^{-7}\,\text{kg},$$
$$\Delta a = \frac{G\,M_{\text{ent}}}{R^2} = \frac{6.674 \times 10^{-11} \times 5.44 \times 10^{-7}}{(10^{-4})^2} = 3.63 \times 10^{-9}\,\text{m/s}^2. \tag{3}$$

The Gemini document claims $3.6 \times 10^{-6}\,\text{m/s}^2$ — the arithmetic contains an error of $\sim 10^3$ somewhere in the unit chain, most likely from treating the distance $R = 100\,\mu\text{m}$ as $100\,\text{m}$ in some intermediate step.

The corrected number $3.6 \times 10^{-9}\,\text{m/s}^2$ still has experimental significance — see Section 5 on experimental redesign.

2.3 Error 3: E_* is a new free parameter, not a repair

The original framework had one free parameter: $\tilde{\kappa}$. After the dimensional fix, there are two: $\tilde{\kappa}$ and E_*. A repair that adds a free parameter is not merely "minimal."

What experiments actually constrain. All experimental observables reduce to the product

$$\Lambda \equiv \tilde{\kappa}\,E_*\,\alpha_{\text{screen}}. \tag{4}$$

An experiment bounding $\mathcal{B}$ measures $\Lambda\,\Delta S_{\text{ent}}^{\text{tot}}/R_{\text{eff}}^2$. The individual values of $\tilde{\kappa}$, E_*, and α_{screen} cannot be separated without additional theoretical input. The paper must state this.

2.4 Fatal concern: The Cooper-pair constraint

This is the most important issue in the entire framework and is absent from every draft to date.

Error

The Cooper-pair constraint. If $E_* \approx E_P$ and $\tilde{\kappa} \approx -1/4$ (unsuppressed), then the predicted "entanglement mass density" of a macroscopic superconductor is orders of magnitude larger than its actual mass. No such gravitational anomaly has been observed. This places a devastating bound on $\tilde{\kappa}E_*$ unless α_{screen} for thermal quantum correlations is essentially zero.

Quantitative estimate. A 1 g niobium superconducting chip has approximately $N_{\text{CP}} \sim 10^{20}$ Cooper pairs. Each Cooper pair in a BCS superconductor is in a two-mode squeezed state with its time-reversed partner, contributing roughly 1 bit of bipartite entanglement across a natural bipartition of the sample. If $E_* = E_P$ and $\tilde{\kappa} = -1/4$:

$$M_{\text{ent}}^{\text{Nb}} \approx N_{\text{CP}} \times m_{\text{bit}} = 10^{20} \times 5.44 \times 10^{-9}\,\text{kg} = 5.4 \times 10^{11}\,\text{kg},$$
$$g_{\text{ent}} \approx \frac{G\,M_{\text{ent}}^{\text{Nb}}}{(1\,\text{mm})^2} \approx 36\,\text{m/s}^2. \tag{5}$$

This is $\sim 5 \times 10^{11}$ times the chip's ordinary gravitational attraction and would be immediately detectable. The non-observation of any such effect places a bound:

$$\tilde{\kappa}E_*\,\alpha_{\text{screen}} \lesssim \frac{\Delta a_{\max}\,R^2}{G\,N_{\text{CP}}} \approx \frac{10^{-12}\,\text{m/s}^2 \times (10^{-3}\,\text{m})^2}{6.67 \times 10^{-11} \times 10^{20}} \approx 10^{-28}\,\text{J/bit}, \tag{6}$$

where $\Delta a_{\max} \sim 10^{-12}\,\text{m/s}^2$ is a conservative estimate of the precision of local gravity measurements.

Resolution options. There are two logically consistent escape routes:

1. $\alpha_{\text{screen}} \to 0$ **for thermal/BCS entanglement.** The internal entanglement entropy S_{int} as defined in the framework (using a coarse-grained bipartition and operational preparation) might be zero for thermally mixed Cooper pairs, because their "entanglement" is mixed-state correlation (classical noise), not pure-state quantum entanglement. This is plausible but must be proven rigorously — it is precisely what makes the S_{int} definition non-trivial.

2. $E_* \ll E_P$ **for all systems.** If $E_* \sim \hbar\omega_{\text{qubit}} \sim 3 \times 10^{-24}\,\text{J}$, then $m_{\text{bit}} \sim 10^{-41}\,\text{kg}$ and no experiment could detect the signal. This is safe but experimentally sterile.

The resolution of the Cooper-pair constraint is the single most important theoretical task. If Option 1 holds rigorously (i.e., the definition of S_{int} excludes all thermal and BCS correlations), then the framework is self-consistent and Option 2 is unnecessary. The paper must prove this or clearly acknowledge it as an open problem.

3 4. Best Theoretical Extension: The EFT Basis

3.1 Cleaned-up action and parameter dictionary

The cleanest form of the action, correcting all errors, is:

$$S[g;\phi] = \int d^4x\,\sqrt{-g}\left[\underbrace{\frac{c^4}{16\pi G}R}_{\text{Einstein-Hilbert}}\quad \underbrace{-\frac{\beta}{2}(\nabla\phi)^2}_{\substack{\text{gradient term}\\ \text{(subleading)}}}\quad \underbrace{-\tilde{\kappa}E_*\phi}_{\text{entanglement source}}\right], \tag{7}$$

with the complete parameter dictionary:

Parameter	Dimensions	Physical meaning
$\phi = S_{\text{ent}}(x)$	$\text{bit}\,\text{m}^{-3} = \text{m}^{-3}$	Internal entanglement entropy density (external source)
$\tilde{\kappa}$	dimensionless	Coupling strength (estimated $-1/4$ under HRH hypothesis)
E_*	J/bit = J	Reference energy per bit (genuinely free parameter)
$\Lambda \equiv \tilde{\kappa}E_*$	J	The only experimentally constrained combination
β	$\text{J}\,\text{m}^5$	Gradient stiffness coefficient (subleading, not yet constrained)
α_{screen}	dimensionless	Environmental screening (0 to 1; to be computed by OP3)
$\Lambda_{\text{eff}} \equiv \tilde{\kappa}E_*\alpha_{\text{screen}}$	J	The only *observable* parameter combination

3.2 Why $\tilde{\kappa}$ and E_* cannot be separated

The action contains only the product $\tilde{\kappa}E_*$ in the linear source term. No other combination of $\tilde{\kappa}$ and E_* appears at leading order ($\beta \to 0$). Therefore:

- Any measurement of a gravitational signal constrains $\Lambda_{\text{eff}} = \tilde{\kappa}E_*\alpha_{\text{screen}}$, not $\tilde{\kappa}$ or E_* separately.
- The "holographic estimate" $\tilde{\kappa} = -1/4$ is only meaningful in combination with a specific hypothesis about E_* (e.g., $E_* = E_P$).
- Papers I–III should always quote bounds on $|\Lambda_{\text{eff}}|$, converting to $|\tilde{\kappa}|$ only when a specific value of E_* is assumed.

3.3 How to think about E_*: the best interpretation

E_* is a **matching coefficient** in an effective field theory. It encodes the unknown UV physics that converts a dimensionless entanglement count (bits) into a physical energy. There are three physically motivated choices:

1. $E_* = E_P$ **(Planck-scale origin, Jacobson-motivated).** Motivation: if entanglement entropy couples to gravity through the same Bekenstein-Hawking mechanism that governs horizon entropy, the natural unit is the Planck energy. The estimate $\tilde{\kappa} = -1/4$ comes from this assumption.
   Consequence: $m_{\text{bit}} = |\tilde{\kappa}| E_P/c^2 \approx 5.4\,\mu\text{g/bit}$. Requires resolving the Cooper-pair constraint via the $\alpha_{\text{screen}} = 0$ argument for uncontrolled thermal correlations.

2. $E_* = \hbar\omega_{\text{qubit}}$ **(device-energy matching).** Motivation: the most natural energy scale for a qubit is its transition energy.
   Consequence: $m_{\text{bit}} \sim 10^{-41}\,\text{kg}$. Completely undetectable with any foreseeable technology.

3. $E_* = k_B T_*$ **for some effective temperature** T_***.** Motivation: Verlinde-style emergent gravity uses a thermal de Sitter temperature.
   Consequence: for $T_* = T_{\text{Unruh}} \sim 10^{-28}\,\text{K}$ (for lab devices), completely undetectable.

Conclusion: Only Option 1 ($E_* \approx E_P$) is experimentally viable with any near-term technology. The paper should adopt this as the benchmark hypothesis while clearly stating it is an assumption, and then bound or refute it experimentally.

3.4 A note on the gradient term

The term $(\beta/2)(\nabla\phi)^2$ carries $[\beta] = \text{J}\,\text{m}^5$ and represents the energy cost of entanglement-density gradients. At the scale of a 1 cm chip with $\nabla\phi \sim S_{\text{ent}}/L \sim 10^6\,\text{m}^{-4}$, this term contributes $(\beta/2)(10^6)^2 = \beta \times 5 \times 10^{11}\,\text{J/m}^3$. For the gradient term to be subleading relative to the $\tilde{\kappa}E_*\phi \sim \tilde{\kappa}E_* S_{\text{ent}} \sim 10^{-1} \times 10^9 \times 10^6 = 10^{14}\,\text{J/m}^3$ (for $E_* = E_P$), we need $\beta \lesssim 6 \times 10^2\,\text{J}\,\text{m}^5$. This is not obviously satisfied for all parameter choices; the subleading assertion must be verified numerically once β is better constrained.

4 5. Mathematical Derivations: Complete and Corrected

4.1 Corrected stress-energy tensor

Varying Eq. (7) with respect to $g^{\mu\nu}$:

$$T^{\text{ent}}_{\mu\nu} = \beta\big(\nabla_\mu\phi\,\nabla_\nu\phi - \tfrac{1}{2}g_{\mu\nu}(\nabla\phi)^2\big) + \tilde{\kappa}E_*\phi\,g_{\mu\nu}. \tag{8}$$

In the quasi-static limit ($\nabla_0\phi \approx 0$), the kinetic term reduces to $-(\beta/2)(\nabla\phi)^2 g_{\mu\nu}$. The leading contribution is $\tilde{\kappa}E_*\phi\,g_{\mu\nu}$ (a cosmological-constant type term with a spatially varying coefficient).

Dimensional check: $[\tilde{\kappa}E_*\phi] = [\text{J/bit}][\text{bit/m}^3] = \text{J/m}^3 = [T_{\mu\nu}]$. ✓

4.2 Exact matter nonconservation (fifth-force law)

From $\nabla^\mu G_{\mu\nu} = 0$ and $G_{\mu\nu} = (8\pi G/c^4)(T^{\text{matter}}_{\mu\nu} + T^{\text{ent}}_{\mu\nu})$:

$$\nabla^\mu T^{\text{matter}}_{\mu\nu} = -\nabla^\mu T^{\text{ent}}_{\mu\nu} = -\big(\tilde{\kappa}E_* + \beta\,\Box\phi\big)\nabla_\nu\phi + \beta\,\nabla_\nu\big(\tfrac{1}{2}(\nabla\phi)^2\big)\,. \tag{9}$$

For a test mass m at rest in a region with an entanglement gradient, the anomalous acceleration is:

$$a^{\nu}_{\text{ent}} = -\frac{1}{\rho_{\text{matter}}}\big(\tilde{\kappa}E_* + \beta\,\Box\phi\big)\nabla^{\nu}\phi. \tag{10}$$

In the quasi-static, slowly varying regime ($|\beta\Box\phi| \ll |\tilde{\kappa}E_*|$, $|\nabla_0\phi| \ll |\nabla_i\phi|$):

$$\mathbf{a}_{\text{ent}} \approx -\frac{\tilde{\kappa}E_*}{\rho_{\text{matter}}}\,\nabla S_{\text{ent}}(\mathbf{x}). \tag{11}$$

Dimensional check: $[\tilde{\kappa}E_*\nabla S_{\text{ent}}/\rho_{\text{matter}}] = [\text{J/m}^3]/[\text{kg/m}^3] = \text{J/kg} = \text{m}^2/\text{s}^2$ per meter $\to$ m/s^2. ✓

4.3 Newtonian limit

The modified Poisson equation for the Newtonian potential Φ:

$$\nabla^2\Phi = 4\pi G\left(\rho_{\text{matter}} + \tilde{\kappa}E_*S_{\text{ent}}\right). \tag{12}$$

Under spherical symmetry:

$$\Delta a(R) = \frac{G\,M_{\text{ent}}(<R)}{R^2}, \qquad M_{\text{ent}}(<R) = 4\pi\int_0^R \tilde{\kappa}E_*S_{\text{ent}}(r)\,r^2\,dr. \tag{13}$$

For a bounded source (radius R_0), this saturates to $\Delta a \propto 1/R^2$ for $R > R_0$.

4.4 The experimentally constrained parameter combination

Any branch-comparison experiment measuring Δa constrains:

$$\boxed{|\Lambda_{\text{eff}}|\cdot\frac{\Delta S^{\text{tot}}_{\text{ent}}}{R^2_{\text{eff}}} \equiv |\tilde{\kappa}_{\text{eff}}|\cdot E_*\cdot\frac{\Delta S^{\text{tot}}_{\text{ent}}}{R^2_{\text{eff}}} = \frac{\Delta a_{\max}\,R^2_{\text{eff}}}{G}} \quad\Rightarrow\quad |\Lambda_{\text{eff}}| < \frac{\Delta a_{\max}\,R^2_{\text{eff}}}{G\,\Delta S^{\text{tot}}_{\text{ent}}}. \tag{14}$$

For the corrected 100-bit, $R = 100\,\mu$m experiment with $\Delta a_{\max} = 10^{-10}$ m/s^2:

$$|\Lambda_{\text{eff}}| < \frac{10^{-10}\times(10^{-4})^2}{6.67\times10^{-11}\times100} = \frac{10^{-18}}{6.67\times10^{-9}} \approx 1.5\times10^{-10}\,\text{J/bit}. \tag{15}$$

Since $E_P = 2\times10^9$ J/bit, this gives $|\tilde{\kappa}\alpha_{\text{screen}}| < 1.5\times10^{-10}/2\times10^9 = 7.5\times10^{-20}$ if $E_* = E_P$. The physical meaning is that even a null result with state-of-the-art force sensing places a nontrivial bound on $|\tilde{\kappa}_{\text{eff}}|$ at the $\sim 10^{-20}$ level when $E_* = E_P$.

5 6. Best Experimental Direction: Optomechanical Force Sensing

5.1 Why force sensing beats Ramsey interferometry

The Ramsey phase shift from a gravitational redshift is suppressed by c^2:

$$\Delta\phi_{\text{Ramsey}} = \omega_{\text{hf}}\,T_{\text{Ramsey}}\,\frac{\Delta\Phi_{\text{ent}}}{c^2} \sim \frac{G\,\Lambda_{\text{eff}}\,\Delta S_{\text{ent}}}{c^2\,R}. \tag{16}$$

For the same parameters, the direct acceleration is:

$$\Delta a = \frac{G\,\Lambda_{\text{eff}}\,\Delta S_{\text{ent}}}{R^2}. \tag{17}$$

The ratio of direct acceleration to Ramsey signal:

$$\frac{\Delta a}{(\Delta\phi/T_{\text{Ramsey}})} = \frac{R\,c^2}{\omega_{\text{hf}}} \approx \frac{10^{-4}\,\text{m} \times 9 \times 10^{16}\,\text{m}^2/\text{s}^2}{2\pi \times 6.8 \times 10^9\,\text{rad/s}} \approx 2 \times 10^5\,\text{m}. \tag{18}$$

The acceleration observable is $\sim 2 \times 10^5$ times larger per unit signal-to-noise ratio than the clock phase shift (for this specific geometry). Force sensing is the correct experimental direction.

5.2 Corrected signal-to-noise analysis

Key Result

Corrected signal (not 3.6×10^{-6} as claimed by Gemini):
$\Delta a = 3.63 \times 10^{-9}\,\text{m/s}^2$ for $E_* = E_P$, $\tilde{\kappa} = -1/4$, $\Delta S_{\text{ent}} = 100\,\text{bits}$, $R = 100\,\mu\text{m}$.

Comparison to state-of-the-art force sensors:

Sensor type	Force sens. [N/$\sqrt{\text{Hz}}$]	Accel. sens. [m/s^2/$\sqrt{\text{Hz}}$]	Signal/noise
Levitated nanosphere ($r = 75\,\text{nm}$)	10^{-20}	5×10^{-3}	7×10^{-7}
Cryogenic micro-cantilever ($m = 1\,\mu\text{g}$)	3×10^{-18}	3×10^{-9}	1.2
MEMS torsion balance ($m = 1\,\mu\text{g}$)	10^{-17}	10^{-8}	0.36
SQUID-based force sensor ($m = 10\,\mu\text{g}$)	10^{-18}	10^{-10}	36

The signal exceeds the noise floor of a cryogenic cantilever by a factor of ~ 1. With a cryogenic SQUID-based force sensor (the most sensitive mechanical force transducer currently demonstrated), the signal exceeds the noise floor by a factor of ~ 36 at 1 Hz bandwidth.

5.3 Recommended flagship experiment: SQUID-lever + superconducting processor

Platform: A superconducting micro-mechanical oscillator (SQUID-readout torsion pendulum) positioned at $R = 100\,\mu\text{m}$ from a 50-qubit fixed-frequency transmon processor.

Protocol:

1. **Branch A (entangled):** Prepare 50-qubit cluster state ($\Delta S^A_{\text{ent}} \approx 25\,\text{bits}$ after realistic α_{screen}).
2. **Branch B (product):** Prepare $|+\rangle^{\otimes 50}$ with identical control schedule but null CZ pulses ($\Delta S^B_{\text{ent}} \approx 0$).
3. **Measure:** Differential force on the lever between branches.
4. **Key systematic:** Differential heat dissipation between branches must be matched to below the force sensitivity level.

Fatal systematic: Differential heating. A 50-qubit gate sequence dissipates $\sim 10^{-8}\,\text{W}$ during circuit execution ($\sim 10^{-6}\,\text{s}$). At $R = 100\,\mu\text{m}$, the differential thermal gradient force (from radiation pressure / phonon-mediated thermomechanical coupling) is $\sim 10^{-13}\,\text{N}$, exceeding the gravitational signal by 10^5.

Mitigation: The circuit must be designed so Branch A and Branch B deposit *identical* heat into the substrate. This requires a matched thermal footprint: the cluster state circuit and the null circuit must have the same total number of physical gate pulses, same pulse amplitudes and durations, and same readout events. Only the CZ interaction phase (the entangling step) differs. This is achievable with careful circuit design.

5.4 Secondary experimental direction: atom interferometry near a chip

A complementary approach places a matter-wave interferometer (e.g., ^{87}Rb Mach-Zehnder) near the superconducting chip and looks for differential phase accumulation between Branch A and B runs. The sensitivity per shot is:

$$\delta(\Delta\phi_{\mathrm{AI}}) = \frac{1}{\sqrt{N_{\mathrm{atoms}}}kT^2}, \tag{19}$$

where k is the photon wavenumber and T the free-fall time. For $N = 10^6$ atoms, $k = 1.6 \times 10^7\,\mathrm{m}^{-1}$, $T = 1\,\mathrm{s}$: $\delta a \approx 3 \times 10^{-14}\,\mathrm{m/s}^2$ per shot. This is 10^5 times more sensitive than the SQUID lever for the same measurement time — but requires the interferometer to operate near a noisy superconducting cryostat, which is technically much harder.

6 7. Paste-Ready Manuscript Text

6.1 Revised theory section (Section 3 replacement)

Manuscript Text

3. COVARIANT FRAMEWORK AND EFFECTIVE ACTION

3.1 Modified field equations. We couple a coarse-grained scalar field $\phi(x) \equiv S_{\mathrm{ent}}(x)$, representing the internal inter-subsystem entanglement entropy density of the quantum device (units: $\mathrm{bit\,m}^{-3}$), to gravity through a minimal effective action. The field ϕ is treated as an externally prescribed source determined by the device quantum state; it is not an independent dynamical degree of freedom.

To convert the dimensionless entropy density to an energy density, we introduce a reference energy scale E_* (J/bit). The effective action is:

$$S[g;\phi] = \int d^4x\,\sqrt{-g}\left[\frac{c^4}{16\pi G}R - \frac{\beta}{2}(\nabla\phi)^2 - \tilde{\kappa}E_*\phi\right],$$

where $\tilde{\kappa}$ is a dimensionless coupling and β (units: $\mathrm{J\,m}^5$) parameterizes gradient corrections. We vary the action with respect to $g^{\mu\nu}$ only; no field equation for ϕ is imposed.

3.2 Entanglement stress-energy tensor. The resulting entanglement contribution to the Einstein equations is:

$$T^{\mathrm{ent}}_{\mu\nu} = \beta\big(\nabla_\mu\phi\,\nabla_\nu\phi - \tfrac{1}{2}g_{\mu\nu}(\nabla\phi)^2\big) + \tilde{\kappa}E_*\phi\,g_{\mu\nu}.$$

In regions where ϕ is slowly varying (gradient term subleading), $T^{\mathrm{ent}}_{\mu\nu} \approx \tilde{\kappa}E_*S_{\mathrm{ent}}\,g_{\mu\nu}$: a spatially varying cosmological constant proportional to the local entanglement entropy density.

3.3 Conservation and fifth force. The contracted Bianchi identity $\nabla^\mu G_{\mu\nu} \equiv 0$ forces:

$$\nabla^\mu T^{\mathrm{matter}}_{\mu\nu} = -\big(\tilde{\kappa}E_* + \beta\,\Box\phi\big)\nabla_\nu\phi + \beta\,\nabla_\nu\big(\tfrac{1}{2}(\nabla\phi)^2\big).$$

In the quasi-static limit, this reduces to a fifth-force law: matter experiences an anomalous force density $-\tilde{\kappa}E_*\nabla S_{\rm ent}(x)$ wherever entanglement density has a gradient. This force is composition-independent but entanglement-dependent, and is therefore not constrained by existing equivalence-principle experiments conducted with classical test masses (for which $S_{\rm ent} \approx 0$).

3.4 Status of $\tilde{\kappa} = -1/4$. The estimate $\tilde{\kappa} = -1/4$ follows from importing Jacobson's Planck-scale area-law regulator into the bulk volume. This step has no rigorous justification; it is a motivated hypothesis. In the absence of a UV derivation, both $\tilde{\kappa}$ and E_* are free parameters. Experiments constrain only their product $\Lambda_{\rm eff} = \tilde{\kappa}E_*\alpha_{\rm screen}$; mapping to $\tilde{\kappa}$ requires additional assumptions about E_*.

6.2 Revised Newtonian limit (Section 5 replacement)

Manuscript Text

5. NEWTONIAN LIMIT AND OBSERVABLE COMBINATION

5.1 Modified Poisson equation. In the Newtonian (weak-field, slow-motion) limit:

$$\nabla^2\Phi = 4\pi G\big(\rho_{\rm matter} + \tilde{\kappa}E_*S_{\rm ent}\big).$$

Under spherical symmetry with enclosed entanglement source $M_{\rm ent}(< R) = 4\pi\int_0^R \tilde{\kappa}E_*S_{\rm ent}(r)\,r^2\,dr$:

$$\Delta a(R) = \frac{G\,M_{\rm ent}(<R)}{R^2} \propto \frac{1}{R^2}, \quad R > R_0.$$

5.2 Experimentally constrained combination. In any branch-comparison experiment, the signal is:

$$\Delta a = \frac{G\,|\Lambda_{\rm eff}|\,\Delta S_{\rm ent}^{\rm tot}}{R_{\rm eff}^2}, \quad \Lambda_{\rm eff} \equiv \tilde{\kappa}E_*\alpha_{\rm screen}.$$

A null result at sensitivity $\Delta a_{\rm max}$ bounds:

$$|\Lambda_{\rm eff}| < \frac{\Delta a_{\rm max}\,R_{\rm eff}^2}{G\,\Delta S_{\rm ent}^{\rm tot}}.$$

For a benchmark: $E_* = E_P = 1.96\times10^9\,\mathrm{J}$, $\tilde{\kappa} = -1/4$, $\alpha_{\rm screen} = 1$ (unsuppressed): $m_{\rm bit} = |\tilde{\kappa}|E_P/c^2 = 5.44\,\mu\mathrm{g/bit}$. For $\Delta S_{\rm ent} = 100\,\mathrm{bits}$ at $R = 100\,\mu\mathrm{m}$: $\Delta a = 3.6\times10^{-9}\,\mathrm{m/s^2}$, within reach of cryogenic force sensors.

7 8. Paper Split

Paper	Core claim	Content	Status
Paper I (4–6 pp, PRD/PRL)	Dimensionally consistent framework with E_*; $1/R^2$ scaling; exact fifth-force formula; correct observable Λ_{eff}	Sections 2–5 of dimensional-fix paper (after c^4 fix); Cooper-pair constraint acknowledged	Ready after micro-edits
Paper II (20–30 pp, PRD)	OP3 Lindblad numerics; OP8 fifth-force bounds; OP10 power counting; E_* interpretation	Full derivations + $\alpha_{\text{screen}}(N, \tau)$ simulation results + bounds from existing systems (Cooper pairs, MEMS, atom interferometry)	Requires OP3 Stage 1–3
Paper III (12 pp, PRA)	SQUID-lever + transmon chip experiment; concrete differential force protocol; systematic error budget	Full experimental proposal, timing, matched-heat systematic analysis	Requires OP3 Stage 3 + thermal simulation

What Paper I needs before submission (2–4 hours of editing):

1. Fix $c^3 \to c^4$ throughout (one search-replace + verify Eq. 5)
2. Add Cooper-pair constraint (one paragraph in the limitations section)
3. Add parameter dictionary table (as above)
4. Change every "$\tilde{\kappa}$" bound to "$\Lambda_{\text{eff}} = \tilde{\kappa} E_* \alpha_{\text{screen}}$" bound
5. Note the Gemini numerical error and use $\Delta a = 3.6 \times 10^{-9}\,\text{m/s}^2$

8 9. Priority Roadmap

Task	Time	Difficulty	Feeds	Status
Fix $c^3 \to c^4$ throughout	30 min	Trivial	Paper I	**Do now**
Write Cooper-pair constraint section	1 day	Moderate	Paper I	**Do now**
Fix Δa numerical value	30 min	Trivial	Paper I + III	**Do now**
Revise all bounds to Λ_{eff}	2 hr	Easy	Paper I	**Do now**
Submit Paper I to arXiv	–	–	Priority	After above
Verify Bose 2023 Nature 623 citation	1 hr	Trivial	All papers	Week 1
OP8: Compute fifth-force from 50-qubit chip	1 day	Moderate	Paper II	Week 1–2
OP10: Power-counting for E_* at lab scales	2 days	Moderate	Paper II	Week 1–2
Prove/disprove $\alpha_{\text{screen}} = 0$ for BCS states	1 week	Hard	Foundational	Month 1
OP3 Stage 1: QuTiP Lindblad $N \leq 10$	2–4 weeks	Moderate	Paper II	Month 1–2
Thermal simulation of chip heat dissipation	1 month	Hard	Paper III	Month 2–3
OP3 Stage 3: MPO $N = 50$	2–4 months	Hard	Paper II + III	Month 3–6

9 10. Hostile Referee Simulation

1. **"The paper introduces E_* as a 'repair' but E_* is a free parameter. The theory now has two undetermined couplings instead of one. How is this an improvement?"**
   Honest answer: The referee is right. The dimensional fix reveals that the original formulation was secretly using units that do not work without an implicit E_*. The repair makes this explicit. The theory is not more predictive after the fix; it is honest about its free parameters. The scientific value is that the framework is now well-posed and experiments can constrain $\Lambda_{\text{eff}} = \tilde{\kappa} E_* \alpha_{\text{screen}}$. Paper I should not overclaim.

2. **"The estimate $\tilde{\kappa} = -1/4$ is stated as 'motivated' but the derivation requires two unjustified assumptions (HRH-1 and HRH-2). Why present this number at all if it has no rigorous status?"**
   Honest answer: $\tilde{\kappa} = -1/4$ is a benchmark value, not a prediction. It provides a concrete reference point for the experimental program. The paper states explicitly that it requires the holographic regulator hypothesis. If this number offends the referee, the paper can present it purely as "if one assumes HRH-1 and HRH-2, then $\tilde{\kappa} = -1/4$." The framework is equally valid for any $\tilde{\kappa}$.

3. **"If $E_* \approx E_P$ (the only experimentally viable choice), then macroscopic superconductors should have anomalous gravitational effects of order 10^5 times their actual gravitational attraction. No such effect is observed. This falsifies the theory."**
   Honest answer: This is the Cooper-pair constraint (Section 3.4) and it is the most important objection. The response is that the framework's S_{int} definition requires pure-state quantum entanglement across a controlled bipartition, not thermal mixed-state correlations. Whether BCS Cooper-pair correlations contribute to S_{int} as defined is a non-trivial open question. If $\alpha_{\text{screen}} = 0$ for all uncontrolled thermal systems (by the operational definition), the constraint disappears. This must be proven, not asserted.

4. **"The action has $c^3/(16\pi G)$ in the Lagrangian. This is dimensionally wrong. The standard EH action uses $c^4/(16\pi G)$."**
   Honest answer: This is a straightforward error that will be corrected in the revised version. The physical results do not change — only the Lagrangian coefficient needs updating. We thank the referee for the catch.

5. **"The paper claims the near-field force experiment has signal $3.6 \times 10^{-6}\,\mathrm{m/s^2}$, but the calculation gives $3.6 \times 10^{-9}\,\mathrm{m/s^2}$. Either the number or the claim about experimental detectability is wrong."**
   Honest answer: The number 3.6×10^{-6} contains an arithmetic error of $\sim 10^3$; the correct value is $3.6 \times 10^{-9}\,\mathrm{m/s^2}$. At this signal level, detection requires a cryogenic SQUID-based force sensor rather than a standard optomechanical setup, but remains achievable with state-of-the-art technology ($\sim 10^{-10}\,\mathrm{m\,s^{-2}\,Hz^{-1/2}}$ for SQUID-readout MEMS). The experimental conclusions are revised but not negated.

References

[1] T. Jacobson, *Phys. Rev. Lett.* **75**, 1260 (1995).

[2] E. Verlinde, *SciPost Phys.* **2**, 016 (2017).

[3] P. Asenbaum *et al.*, *Phys. Rev. Lett.* **125**, 191101 (2020).

[4] P. T. Touboul *et al.*, *Phys. Rev. Lett.* **129**, 121102 (2022).

[5] S. Bose *et al.*, *Phys. Rev. Lett.* **119**, 240401 (2017).

[6] U. Delić *et al.* [levitated nanoparticle force sensing], *Science* **367**, 892 (2020).

[7] D. Rugar *et al.* [SQUID-cantilever force sensing], *Science* **264**, 1560 (1994).

[8] R. Blume-Kohout *et al.*, *Nat. Commun.* **8**, 14485 (2017).